Feasibility Study On Churn Prediction for Nordic Market

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1. Introduction

Our mission is to help clients in Healthcare & Citizen Welfare, Industry & Trade Management, and Construction & Property Management reduce customer attrition by leveraging advanced churn prediction models. By tailoring our offerings to each vertical's unique data signals and business processes, we empower organizations to identify at-risk users early and take proactive retention actions.

2. Defining Churn by Vertical

Vertical	Definition of Churn
Healthcare & Citizen Welfare	Practitioner or patient who ceases platform use or subscription within a rolling period (e.g., 60 days without login/appointment).
Industry & Trade Management	Company or user entity that discontinues transaction activity or license renewal within the contract term.
Construction & Property Management	Builder, tenant, or manager that does not renew service agreements or fails to submit projects/leases over a defined window.

3. Core Questions Our Churn Model Answers

1. **Who is at highest risk of churn?**
2. **Which behaviors/events precede churn?**
3. **What timing/window is critical for intervention?**
4. **Which customer segments should we prioritize?**

5. **What retention actions yield the best ROI?**

4. Data Requirements & Key Data Points

4.1 Customer Profile & Demographic Data

- Organization size, specialty/industry type
- Role (doctor, builder, tenant, trade partner)
- Contract type & start/end dates

4.2 Usage & Engagement Metrics

- Login frequency, session duration, feature utilization
- Number of transactions or projects processed
- Support tickets opened, response times

4.3 Financial & Billing Signals

- Payment history, invoice lag
- Subscription upgrades/downgrades
- Discounts applied or negotiated terms

4.4 Interaction & Feedback Data

- Survey responses (CSAT, NPS)
- Training attendance, certification completions
- Communication logs (emails, calls)

4.5 External & Contextual Factors

- Market/regulatory changes (e.g., new healthcare compliance)
- Economic indicators (construction activity index, trade volumes)

5. Our Offering & Implementation Roadmap

1. **Data Integration**: Seamless connectors to ERP, CRM, EMR, and billing systems.
2. **Feature Engineering**: Domain-specific metrics (e.g., patient no-show rates, project milestones missed).
3. **Model Development**: Ensemble algorithms calibrated per vertical with explainable outputs.

4. **Dashboard & Alerts**: Real-time risk scoring, segment filters, and action recommendations.
5. **Retention Playbooks**: Automated email/SMS campaigns, in-app prompts, and account manager task lists.

6. Business Impact & ROI

- **Increase Renewal Rates** by up to 15% through targeted interventions.
- **Reduce Support Costs** by preempting issues before escalation.
- **Boost Lifetime Value** via upsell opportunities for engaged, low-risk users.

Ready to transform churn into opportunity? Let's schedule a workshop to map your data, define KPIs, and pilot a proof-of-concept in your top-priority vertical.

Understanding Churn Prediction Across Different Verticals

Let me help you develop a churn prediction pitch customized for your three vertical software products. Each domain will require specific approaches since the definition of "churn" and relevant indicators vary by industry and user type.

Definition of Churn Prediction by Vertical

Healthcare and Citizen Welfare

For healthcare, churn prediction means identifying which healthcare providers (doctors, clinics, hospitals) or citizen welfare organizations are likely to discontinue using your software solutions.

Industry and Trade Management

In this vertical, churn prediction involves forecasting which manufacturing companies, trading businesses, or supply chain partners may stop using your trade management platforms.

Construction and Property Management

Here, churn prediction focuses on identifying which builders, property developers, or property management companies might discontinue your construction or tenant management solutions.

Key Questions Your Churn Prediction Model Will Answer

1. **Which customers are at risk of churning in the next 3-6 months?**

2. **What are the early warning signs of potential churn for each vertical?**
3. **What specific factors most strongly correlate with churn in each domain?**
4. **What targeted interventions could prevent high-value customers from churning?**
5. **How does churn vary by user type within each vertical (e.g., doctors vs. administrators)?**
6. **What is the expected ROI from churn prevention activities?**

Required Data Points by Vertical

Healthcare and Citizen Welfare

- **Usage metrics**: Frequency of login, features used, time spent in system
- **User satisfaction**: NPS scores, support ticket frequency/severity
- **Contract details**: Renewal dates, contract value, payment history
- **User demographics**: Facility size, specialty, number of practitioners
- **System adoption**: Percentage of potential users actively using the system
- **Competitor activities**: Recent competitor outreach, market changes

Industry and Trade Management

- **Transaction volume**: Changes in processing volumes
- **Integration depth**: Number of connected systems/processes
- **Business performance**: Customer's business growth/contraction
- **Seasonal patterns**: Expected vs. actual usage during key periods
- **Feature utilization**: Advanced vs. basic feature adoption
- **Support engagement**: Training participation, knowledge base usage

Construction and Property Management

- **Project lifecycles**: Project completion rates, new project starts
- **Property portfolio changes**: Growth or reduction in managed properties
- **Seasonal usage patterns**: Peak vs. off-peak engagement
- **User roles engagement**: Builder vs. tenant manager activity levels
- **Mobile vs. desktop usage**: Field vs. office utilization
- **Regulatory compliance utilization**: Usage of compliance-related features

Documentation for Churn Prediction Pitch

To create a compelling pitch for your churn prediction offering, I recommend structuring your documentation as follows:

1. **Executive Summary**
    - High-level overview of the churn prediction solution
    - Expected business impact and ROI
2. **Vertical-Specific Churn Definitions**
    - Custom definitions of what constitutes churn in each domain
    - Why traditional churn models don't work across these specialized industries
3. **Data Requirements and Collection**
    - Required data points (as outlined above)
    - Integration requirements with existing systems
    - Data privacy and security considerations (especially critical for healthcare)
4. **Prediction Methodology**
    - Machine learning approach tailored to each vertical
    - Explanation of how the model handles domain-specific factors
5. **Actionable Outputs**
    - Risk scoring system
    - Early warning alerts
    - Recommended intervention strategies by customer segment
6. **Implementation Roadmap**
    - Timeline for deployment
    - Integration with existing customer success workflows
    - Training requirements for your team
7. **Success Metrics**
    - How to measure the effectiveness of the churn prediction model
    - Benchmarks for success in each vertical

This structured approach will help you demonstrate the value of a churn prediction system that understands the unique characteristics of each vertical rather than applying a one-size-fits-all solution.

Revolutionizing Customer Retention: A Predictive Churn Solution for Your Vertical Software Portfolio

The Challenge: Silent Erosion of Your Customer Base

In today's competitive software landscape, particularly within specialized Nordic markets, retaining hard-won customers is paramount. Losing a customer isn't just a single lost subscription; it's a loss of recurring revenue, potential up-sell opportunities, and valuable market feedback. For our distinct product divisions – Healthcare and Citizen Welfare, Industry and Trade Management, and Construction and Property Management – understanding and proactively addressing customer churn is critical for sustained growth and profitability.

Introducing Predictive Churn Management: From Reactive to Proactive

We propose the development of a sophisticated **Customer Churn Prediction** solution tailored to the unique dynamics of each of your vertical software offerings. This isn't a one-size-fits-all approach; it's a targeted strategy that recognizes the different indicators of churn across diverse user bases – from doctors utilizing healthcare platforms to builders managing construction projects and tenant managers overseeing properties.

Defining Churn in Your Verticals: A Tailored Perspective

At its core, **customer churn** refers to the event when a customer ceases to use your software product or service and discontinues their subscription or relationship with your company. However, the *manifestation* and *leading indicators* of churn will differ significantly across your divisions:

- **Healthcare and Citizen Welfare:**
    - **For Doctors/Clinics:** Churn could mean a clinic discontinuing their subscription to your Electronic Health Record (EHR) system, practice management software, or telehealth platform. This might be indicated by a significant drop in patient record entries, appointment scheduling, or logins by medical staff. It could also be a failure to renew an annual license or a formal contract termination.
    - **For Citizen Welfare Portals:** Churn might be defined as a municipality or social service agency ceasing to use your platform for case management or citizen communication, or a significant decline in active users or submitted applications.

- **Industry and Trade Management:**
    - **For Businesses/Users:** Churn would signify a company stopping their use of your supply chain management, inventory control, or trade documentation software. Indicators could include a sharp decrease in transaction volumes, data uploads, user logins from key personnel, or the non-renewal of service agreements. For specific modules, churn could mean a cessation of use of that particular functionality.
- **Construction and Property Management (Nordics):**
    - **For Builders/Construction Companies:** Churn could be defined as a construction firm no longer using your project management, BIM (Building Information Modeling), or estimation software for new projects. This might be signaled by a lack of new project creation, a drop in collaboration activities on the platform, or failure to renew licenses post-project completion.
    - **For Tenant Management/Property Owners:** Churn could mean a property management company or landlord discontinuing the use of your tenant communication, rent collection, or maintenance tracking software. Indicators might include a halt in new tenant onboarding, a decrease in processed payments, or a lack of engagement with communication features.

Key Questions Our Churn Prediction Model Will Answer:

Our predictive model will provide actionable insights by answering critical questions such as:

1. **Who is at risk?** Which specific customers (doctors, clinics, businesses, builders, property managers) are most likely to churn in the near future (e.g., next 30, 60, or 90 days)?
2. **Why are they at risk?** What are the key factors and patterns (e.g., declining software usage, specific unutilized features, unresolved support tickets, changes in user roles, payment issues) that indicate a high churn probability for different customer segments within each vertical?
3. **When are they likely to churn?** Can we predict the timeframe within which a high-risk customer is likely to discontinue service?
4. **What is the financial impact?** What is the potential revenue loss associated with the predicted churn?

5. **How can we intervene?** What targeted interventions (e.g., proactive support, specialized training, feature highlighting, tailored offers) are most likely to retain at-risk customers based on their specific churn drivers and user profile?
6. **How do churn drivers differ across verticals and user types?** For example, are the reasons doctors churn different from why construction builders churn? Understanding these nuances is key to effective intervention.
7. **What product features (or lack thereof) are correlated with churn?** Are customers who don't use a specific new module more likely to churn?
8. **How effective are our retention strategies?** By tracking interventions, we can measure their success in reducing churn.

Data Points: The Foundation of Predictive Power

To build an accurate and robust churn prediction model, we will require access to a variety of data points. The specific data will be tailored to each vertical, but generally includes:

I. Common Data Points (Across All Verticals):

- **Customer Account Information:**
    - Customer ID
    - Company Name / User Name
    - Subscription Plan/Tier
    - Contract Start and End Dates
    - Billing Information (payment history, failed payments)
    - Customer Demographics (e.g., size of clinic/company, location - relevant for Nordics focus)
    - Signup Date / Customer Tenure
- **Product Usage & Engagement:**
    - Login Frequency & Recency
    - Session Duration
    - Key Feature Usage (specific to each product – e.g., number of patient records accessed, projects created, invoices sent)
    - Module Adoption Rate
    - Number of Active Users (for company accounts)

    - Frequency of specific actions (e.g., reports generated, data imported/exported)
    - Error Rates Encountered by Users
- **Customer Support & Interaction Data:**
    - Number of Support Tickets Raised
    - Support Ticket Severity & Resolution Time
    - Sentiment from Support Interactions (if available)
    - Participation in Webinars/Training
    - NPS (Net Promoter Score) / CSAT (Customer Satisfaction) Scores
- **Marketing & Communication Engagement:**
    - Email Open Rates & Click-Through Rates
    - Engagement with Newsletters or Product Updates

II. Vertical-Specific Data Point Examples:

- **Healthcare and Citizen Welfare:**
    - **Doctors/Clinics:**
        - Number of patient appointments scheduled/managed
        - Volume of prescriptions issued/managed via the system
        - Integration usage with lab or imaging systems
        - Compliance reporting activity
    - **Citizen Welfare Portals:**
        - Number of cases managed/closed
        - Citizen interaction rates (messages, applications)
        - Workflow completion rates
- **Industry and Trade Management:**
    - Volume of transactions processed (orders, shipments, invoices)
    - Inventory levels managed
    - Number of trade documents generated/managed
    - User role activity (e.g., finance users vs. logistics users)

    - Integration usage with ERP or accounting systems
- **Construction and Property Management (Nordics):**
    - **Builders/Construction Companies:**
        - Number of active projects managed
        - Frequency of BIM model updates/collaboration
        - Volume of tenders or bids created/managed
        - Mobile app usage on-site
    - **Tenant Management/Property Owners:**
        - Number of properties/units managed
        - Rent collection success rates
        - Maintenance request volume and resolution times
        - Tenant communication logs (volume, response times)

Documentation for Churn Prediction Pitch:

To effectively pitch this solution, we will develop the following documentation:

1. **Executive Summary:** Briefly outlining the problem, proposed solution, and expected benefits.
2. **The Problem of Customer Churn:** Detailing the impact of churn specifically for the company and its three divisions.
3. **Our Proposed Solution: Predictive Churn Management:**
    - Tailored Churn Definitions for each vertical and user type.
    - Overview of the predictive modeling approach.
    - Key Questions the model will answer.
4. **Data Requirements:**
    - Comprehensive list of required common and vertical-specific data points.
    - Data sources and accessibility.
    - Data quality considerations.
5. **Expected Benefits & ROI:**
    - Increased customer retention rates.
    - Reduced revenue loss.

    - Improved customer satisfaction and loyalty.
    - Actionable insights for product development and service improvement.
    - More efficient resource allocation for retention efforts.
6. **Implementation Roadmap & Timeline (High-Level):**
    - Phase 1: Data Collection & Preparation
    - Phase 2: Model Development & Validation (per vertical)
    - Phase 3: Pilot Program & Deployment
    - Phase 4: Monitoring & Iteration
7. **Required Resources & Next Steps:**
    - Cross-functional team involvement (data science, product, sales, support).
    - Technology stack considerations.
    - Call to action and proposed next steps.

By investing in this tailored Churn Prediction solution, we can empower each product division to proactively identify and retain valuable customers, fostering sustainable growth and solidifying our market leadership in the Nordics.

Predicting and Preventing Customer Departure: A Proactive Approach to SaaS Growth

In today's competitive SaaS landscape, retaining existing customers is paramount to sustainable growth. Customer churn, the rate at which customers discontinue their subscriptions, poses a significant threat to revenue and profitability. This pitch outlines a data-driven approach to predict and mitigate customer churn across our diverse vertical software products, specifically tailored for Healthcare and Citizen Welfare, Industry and Trade Management, and Construction and Property Management in the Nordics.

Our churn prediction initiative aims to empower the company to proactively identify at-risk customers, understand the root causes of their potential departure, and implement targeted retention strategies. By leveraging data unique to each domain and user type – be it doctors utilizing specialized healthcare modules, builders managing complex construction projects, or tenants interacting with property management portals – we can develop a nuanced understanding of churn drivers.

Defining Churn: A Tailored Perspective

At its core, **customer churn** for our vertical software products signifies the cessation of a customer's paid subscription or the non-renewal of a contract for a specific product or service. However, the *meaning* of churn and its early indicators can differ significantly based on the vertical and the specific user:

- **Healthcare and Citizen Welfare:**
    - **For Doctors/Clinics:** Churn could mean a clinic discontinuing its subscription to our Electronic Health Record (EHR) system, practice management software, or a specialized diagnostic tool. Early indicators might include a significant drop in daily active users, reduced usage of core features (e.g., appointment scheduling, e-prescribing), a surge in support tickets related to usability, or non-engagement with new feature rollouts.
    - **For Citizen Welfare Agencies:** Churn might involve a government agency or department ceasing to use our case management or citizen portal software. Indicators could be a decline in new case entries, low adoption rates of specific modules designed for welfare program management, or failure to renew a multi-year service agreement.
- **Industry and Trade Management:**
    - **For Businesses/Traders:** Churn would be defined as a company cancelling its subscription to our Enterprise Resource Planning (ERP), supply chain management, or trade documentation software. Precursors could include decreased transaction volumes, infrequent logins by key personnel, failure to integrate new trading partners into the system, or complaints about system performance during peak trading hours.
- **Construction and Property Management (Nordics):**
    - **For Builders/Construction Companies:** Churn could signify a construction firm ending its use of our project management, Building Information Modeling (BIM), or estimation software. Warning signs might be a drop in new project setups, underutilization of collaborative features, a decline in data uploads (e.g., blueprints, progress reports), or a switch to a competitor's platform for new tenders.
    - **For Tenant Management/Property Owners:** Churn could mean a property management company or owner stops using our tenant communication, rent collection, or maintenance request tracking software. This could be signaled by a low volume of tenant interactions through the platform, a high number of unresolved maintenance tickets,

or the decision to manage properties through alternative, perhaps less sophisticated, means.

Key Questions Our Churn Prediction Model Will Answer

By implementing a robust churn prediction model, we aim to answer critical business questions, enabling proactive interventions:

1. **Who is likely to churn?** Identifying specific customers (doctors, clinics, businesses, builders, property managers) at high risk of discontinuing their subscription in the near future (e.g., next 30, 60, or 90 days).
2. **Why are they likely to churn?** Understanding the key drivers and factors contributing to churn risk for different customer segments and user types within each vertical (e.g., low product adoption, unmet feature needs, poor user experience, pricing concerns, external market factors specific to the Nordics).
3. **When are they likely to churn?** Predicting the timeframe within which a customer is expected to churn, allowing for timely and prioritized intervention efforts.
4. **What is the financial impact of potential churn?** Quantifying the potential revenue loss associated with at-risk customers to prioritize retention efforts on high-value accounts.
5. **Which intervention strategies are most effective?** Determining the most impactful actions (e.g., targeted training, proactive support, special offers, feature enhancements) to retain specific at-risk customer segments based on their churn reasons.
6. **How do product usage patterns correlate with churn?** Identifying specific features or usage thresholds that are strong indicators of customer stickiness or, conversely, churn risk (e.g., are doctors who don't use the e-referral feature more likely to churn? Are construction companies that don't utilize the daily reporting module at higher risk?).
7. **What are the differences in churn drivers across our Nordic verticals?** Understanding the unique challenges and customer expectations in Healthcare, Industry & Trade, and Construction & Property Management that influence churn.

Essential Data Points for Churn Prediction

To build an effective churn prediction model, we will require a comprehensive set of data points. This data will be documented and managed with strict adherence to data privacy and security regulations, especially concerning sensitive healthcare information.

I. Customer Account & Subscription Data:

- **Customer ID/Account ID:** Unique identifier for each customer.
- **Company Name & Size:** (Especially for B2B contexts in Industry/Trade and Construction).
- **Subscription Plan Details:** Current plan, plan history, pricing, contract start and end dates, renewal dates.
- **Billing Information:** Payment history, overdue payments, payment method.
- **Contract Length & Terms:** Duration of the agreement, any specific clauses.
- **Customer Tenure:** How long they have been a customer.
- **Geographic Location:** (Nordics focus, potentially specific country data if relevant).
- **Vertical/Sub-vertical:** Specific segment within Healthcare, Industry/Trade, or Construction/Property.
- **User Roles & Permissions:** (e.g., admin users, standard users for doctors, project managers, tenants).

II. Product Usage & Engagement Data:

- **Login Activity:** Frequency of logins, recency of last login, session duration for different user types.
- **Feature Adoption & Usage:** Which features are being used, how often, depth of feature usage (e.g., number of reports generated, number of patient records accessed, number of trade documents processed, number of construction projects managed, number of tenant communications sent).
- **Activity Metrics:** Number of key actions performed (e.g., appointments booked, invoices created, projects updated, maintenance requests logged).
- **User Engagement Score:** A composite score based on various interaction metrics.
- **License Utilization:** Number of active users vs. purchased licenses.
- **Data Volume:** Amount of data stored or processed (e.g., patient records, project files, transaction data).
- **Integration Usage:** Use of APIs or integrations with other software.
- **Mobile App Usage (if applicable):** Engagement and feature usage on mobile platforms.

- **Last Activity Date on Key Features:** Identifying dormancy for critical product functionalities.

III. Customer Interaction & Support Data:

- **Support Ticket History:** Number of tickets, ticket severity, resolution time, ticket topics (e.g., bugs, feature requests, usability issues).
- **Customer Satisfaction (CSAT) Scores:** Feedback from surveys post-support interaction.
- **Net Promoter Score (NPS):** Overall customer loyalty and satisfaction.
- **Communication History:** Records of interactions with sales, account management, and customer success teams (e.g., emails, calls, meeting notes).
- **Training & Onboarding Data:** Participation in training sessions, completion of onboarding milestones.
- **Community Forum Activity:** Engagement in user forums or communities.
- **Website & Help Center Usage:** Pages visited, searches performed.

IV. Vertical-Specific Data Considerations:

- **Healthcare and Citizen Welfare:**
    - Adoption rates of specific clinical modules (e.g., e-prescription, lab integration for doctors).
    - Number of patient encounters managed (for clinics).
    - Compliance-related feature usage.
    - System uptime and performance metrics (critical for healthcare).
- **Industry and Trade Management:**
    - Transaction volumes and values processed.
    - Number of trading partners connected or managed.
    - Inventory turnover rates (if applicable to the software).
    - Usage of regulatory compliance features.
- **Construction and Property Management (Nordics):**
    - Number of active projects managed (for builders).
    - Frequency of project updates and collaboration activities.
    - Tenant onboarding success rates (for property management).

    - Volume of maintenance requests processed and resolution times.
    - Usage of specific Nordic-related compliance or reporting features.

V. Feedback & Sentiment Data:

- **In-app Feedback:** Direct feedback submitted through the software.
- **Surveys:** Responses to churn surveys, product satisfaction surveys.
- **Social Media Mentions (if applicable and monitorable):** Public sentiment.